

1 Static Yao-Kivelson Model

1.1 Kitaev Model and Yao-Kivelson Model

Kitaev found a way to map the strongly-interacting spin model

$$H = - \sum_{\substack{\text{x-bonds} \\ \langle jk \rangle}} J_{jk}^x \sigma_j^x \sigma_k^x - \sum_{\substack{\text{y-bonds} \\ \langle jk \rangle}} J_{jk}^y \sigma_j^y \sigma_k^y - \sum_{\substack{\text{z-bonds} \\ \langle jk \rangle}} J_{jk}^z \sigma_j^z \sigma_k^z, \quad (1.1)$$

now known as the Kitaev Model, to a free Majorana-Fermion hopping model, and therefore managed to solve the model exactly [1]. Classically, 'the ' Kitaev model is defined on a honeycomb lattice, where each coupling type $\alpha = x, y, z$ corresponds to one of the three bond orientations. However, his method directly generalizes to any tricoordinate lattice geometry [2], provided that each site has exactly one bond of each type $\alpha = x, y, z$.

One alternative to the honeycomb lattice is the star/triangle-honeycomb lattice. Instead of elementary plaquettes of size 6, it features two different types of plaquettes of size 3 and 12, respectively. When defined on this geometry, model 1.1 is known as the Yao-Kivelson Model [3]. Due to the odd length of the triangular plaquettes, this model features a groundstate that spontaneously breaks Time-Reversal Symmetry, and therefore describes a Chiral Spin Liquid.

1.2 Kitaev's Spin-Majorana Mapping

Kitaev introduced a representation of a spin- $\frac{1}{2}$ -degree of freedom (called 'spin' in the following) in terms of 4 fermionic Majorana modes. Consider the 4-dimensional Fockspace $\mathcal{H}$ of two independent abstract fermions a_1, a_2 satisfying the canonical anti-commutation relations

$$\left\{ a_k, a_l^\dagger \right\} = \delta_{kl}, \quad \left\{ a_k, a_l \right\} = 0. \quad (1.2)$$

They give rise to 4 real fermion modes

$$c_{2k-1} := c_k + c_k^\dagger \quad \text{and} \quad c_{2k} := \frac{1}{i} \left(a_k - a_k^\dagger \right), \quad \text{for} \quad k = 1, 2, \quad (1.3)$$

which are commonly called Majorana modes. They satisfy

$$c_i^\dagger = c_i \quad \text{and} \quad \{c_i, c_j\} = 2\delta_{ij}, \quad (1.4)$$

which qualifies them to be called "real fermions", in contrast to the usual complex ones.

The goal is now to use these to represent the spin algebra

$$\sigma^\alpha \sigma^\beta = \delta^{\alpha\beta} + i\epsilon^{\alpha\beta\gamma} \sigma^\gamma, \quad \alpha, \beta, \gamma = x, y, z. \quad (1.5)$$

For that, rename the 4 Majorana modes from $\{c_i\}_{i=1,\dots,4}$ to $\{b^x, b^y, b^z, d\}$ and define

$$\sigma^\alpha := ib^\alpha d, \quad \alpha = x, y, z. \quad (1.6)$$

A necessary condition for eq. 1.5 to hold is $\sigma^x \sigma^y \sigma^z = i$, which implies

$$1 \stackrel{!}{=} b^x b^y b^z d =: D, \quad (1.7)$$

where we have defined the hermitean operator D , which cannot be the identity on $\mathcal{H}$, as it anti-commutes with each Majorana operator:

$$\{D, c_i\} = 0 \quad \text{for} \quad c_i = b^x, b^y, b^z, d. \quad (1.8)$$

Due to $D^2 = 1$, $\mathcal{H}$ splits into the eigenspaces $\mathcal{H} = \mathcal{H}_1 \oplus \mathcal{H}_{-1}$, where the subscript denotes the respective eigenvalues of D . Consequently, $\mathcal{H}_1$ is the space where eq. 1.7 is fulfilled. Each Majorana operator c_i defines a bijection

$$c_i : \mathcal{H}_1 \rightarrow \mathcal{H}_{-1}, \quad |\psi\rangle \mapsto c_i |\psi\rangle \quad \text{with} \quad c_i^{-1} = c_i, \quad (1.9)$$

which directly follows from eq. 1.8.

of 2 fermion modes on a 4-dimensional Hilbertspace $\mathcal{H}$, giving rise to 4 Majorana Fermion modes $\{b^x, b^y, b^z, d\}$. On the Fock-Space $\mathcal{F}$

Defining the hamiltonian 1.1 on a

Bibliography

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